

Online supplement: The Fiscal Incidence of Trade-Related Labour-Income Shocks in the UK

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A Data appendix

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55,600 million customs-basis total, £52,500 million maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

B Shock mechanics

Each employee in exposed division j is independently selected with probability s_j . Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove s_j of each exposed worker’s earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

C Monte Carlo and measures

Central direct and measured scenarios use 100 assignments. Every displayed $\pm$ value is an assignment SD, not a standard error or confidence interval. The machine-readable artifacts additionally store numerical Monte Carlo standard error as assignment SD divided by $\sqrt{n}$; it measures precision of the simulated mean only. Survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment take-up, mechanism, demographic, poverty-gap, reallocation, duration and sensitivity-grid exercises, and the ten-assignment supply-chain exercise, are exploratory and are excluded from the headline evidence. The mixed-adjustment scenario surface likewise uses five common assignments per cell. Its machine-readable cells and draws are in `results/scenario_testing.json` and `results/scenario_testing.csv`.

Reported relative BHC poverty changes hold the poverty line frozen at 60 per cent of the baseline equivalised median; absolute BHC poverty likewise uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is $1 - \Delta Y / \Delta E$, where ΔE is gross employment-earnings loss and ΔY is household disposable-income loss.

Reconciling cushioning and the Exchequer effect. The Exchequer effect is the change in the aggregate simulated government balance, not the household-side transfer implied by the cushioning identity. The identity implies a household-side offset of $(1 - \Delta Y / \Delta E) \times \Delta E = \Delta E - \Delta Y$: for the full-schedule wage cut, $0.439 \times \text{£}886\text{m} \approx \text{£}389$ million, and for displacement $0.368 \times \text{£}882\text{m} \approx \text{£}325$ million, against reported Exchequer costs of £484 million and £412 million. The wedge is revenue lost outside employee disposable income: employer National Insurance contributions fall with the employee wage bill under every margin, and for displaced workers the zeroing of employee pension contributions and salary-sacrifice arrangements changes taxable pay relative to the gross-earnings accounting in ΔE . The government balance also

absorbs any other simulated fiscal flows that respond to the shock. $\text{Cushion} \times \text{gross}$ therefore understates the Exchequer cost by roughly £87–95 million in the central full-schedule scenarios, and the two statistics answer different questions: the cushioning rate measures household income protection per pound of employee earnings lost, while the Exchequer effect measures the total first-round cost to the public finances.

D Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{0.4, 1, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

E Factorial adjustment scenarios

Figure 1 separates the aggregate export-demand calibration from the assumed composition of labour-market adjustment. Each cell uses the full tariff schedule and five common assignments. The wage-cut endpoint is stable: cushioning is 40.6–41.3 per cent over the four calibrations. Moving towards displacement generally lowers cushioning, but the small-shock cells are especially noisy because a few high-weight FRS records dominate five Bernoulli assignments. These cells are stress scenarios, not estimates or confidence regions. The qualitative result is that fiscal cost scales mainly with the imposed gross shock, while the adjustment mix changes the share absorbed by taxes and transfers.

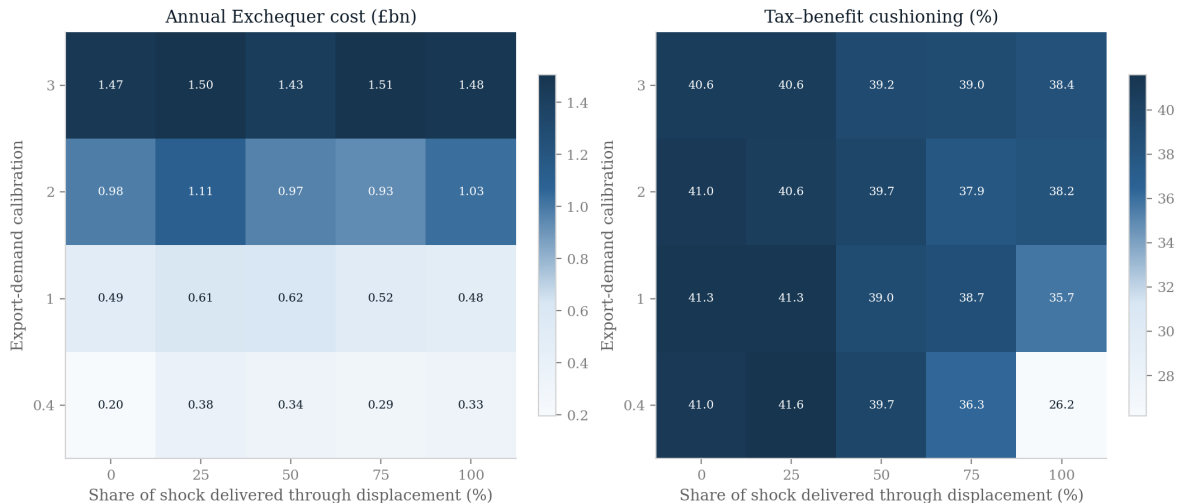


Figure 1: Factorial scenario surface: export-demand calibration by the share of expected sector loss delivered through displacement rather than survivor wage cuts. Cell values are means over five common assignments. Assignment noise, survey uncertainty and uncertainty in the production-side transmission are not combined.

F Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £264 million, Exchequer cost of $£149 \pm 131$ million, cushioning of 39.7 ± 3.4 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £432 million and the Exchequer cost to $£245 \pm 216$ million. The 14 per cent penalty gives gross loss of £131 million and Exchequer cost of $£75 \pm 64$ million. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers' penalty is removed.

G Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 36.6 ± 4.9 per cent and costs the Exchequer $£419 \pm 228$ million. The wage-cut scenario is cushioned by 44.1 per cent and costs £517 million. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

H HMRC destination-panel diagnostic

The machine-readable specification, clustered standard errors, individual destination checks, sample-start sensitivities, tariff-intensity groups and May 2019–23 pseudo-policy estimates are stored in the event-study JSON artifact; the plotted observations are stored in its companion monthly CSV. Pre-policy value weights are capped at the 99th percentile, giving an effective product count of 107 despite 1194 included HS4 products. Uncapped and equal-weight estimates disclose the remaining concentration sensitivity.

I Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

J Duration scope

The submission design reports three-, six- and twelve-month *duration-equivalent annual stresses*. It scales the probability of a coherent full-year displaced state rather than combining partial-year earnings with full-year unemployment status. These rows preserve expected worker-month loss but are not unemployment-spell simulations. Actual within-year tax, UC, waiting-period and re-employment timing requires a monthly model and is not claimed here.

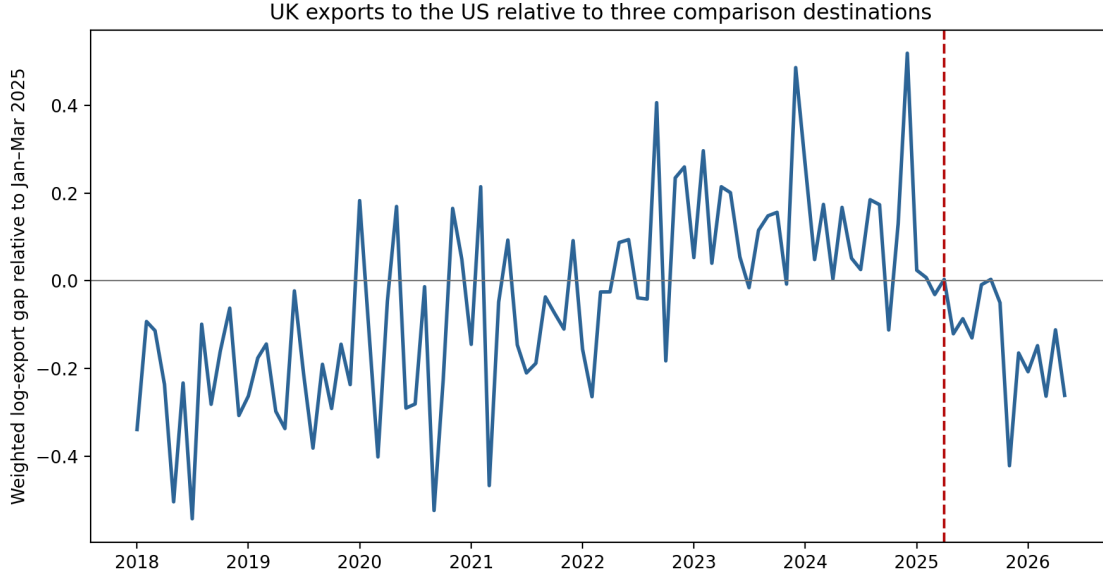


Figure 2: Monthly capped-value-weighted gap between UK HS4 exports to the United States and mean exports of the same products to Canada, Japan and Australia, normalised to January–March 2025. The dashed line marks April 2025, which is omitted from the regression as an announcement and anticipation month. The series is descriptive and has no causal confidence band.

K Secondary distributional and fiscal figures

L Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory ten-assignment microsimulation, displacement affects 29,700 weighted workers on average, costs the Exchequer $\pounds 680 \pm 310$ million and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs $\pounds 960$ million and leaves measured relative poverty unchanged. These figures are accounting sensitivities, not all-channel causal estimates.

M Constituency illustration

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. It is averaged over 20 paired worker-assignment draws (seeds 0–19), replacing the unstable single-seed surface used in an earlier draft. The mapping now divides each household’s disposable-income change by household size before constructing constituency means. The CSV reports both the draw mean and assignment-draw standard deviation. Figure 6 shows the resulting synthetic surface. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, these remain synthetic demographic and income projections, not constituency exposure estimates; individual seat rankings are therefore deliberately

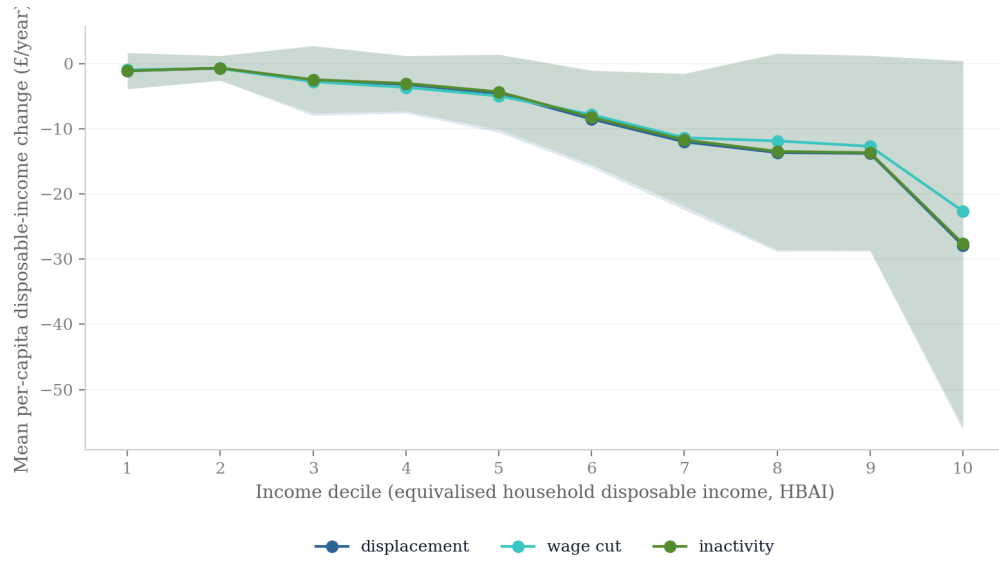


Figure 3: Per-capita disposable-income change by baseline income decile in the legacy full-schedule scenarios. Shading is assignment dispersion.

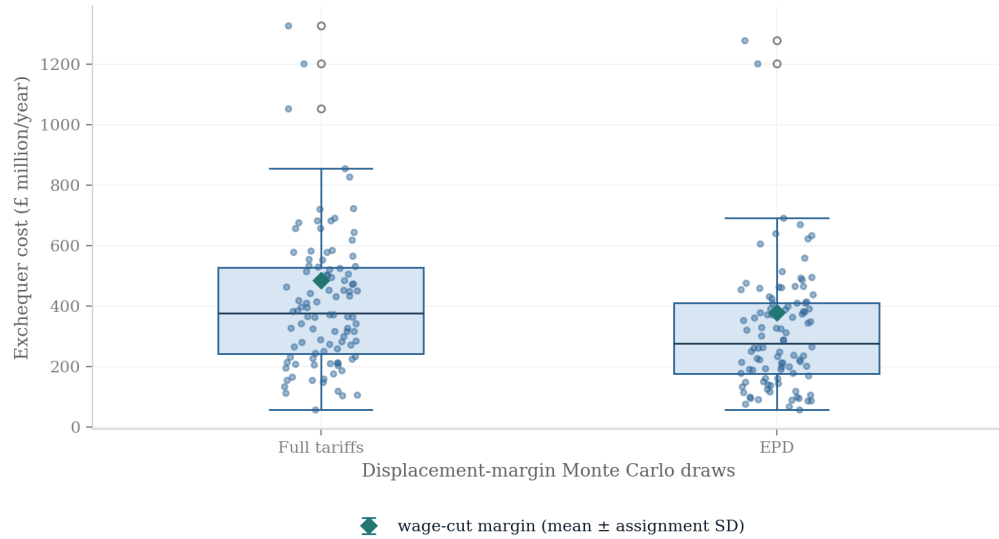


Figure 4: Annual Exchequer cost across legacy regenerated assignments.

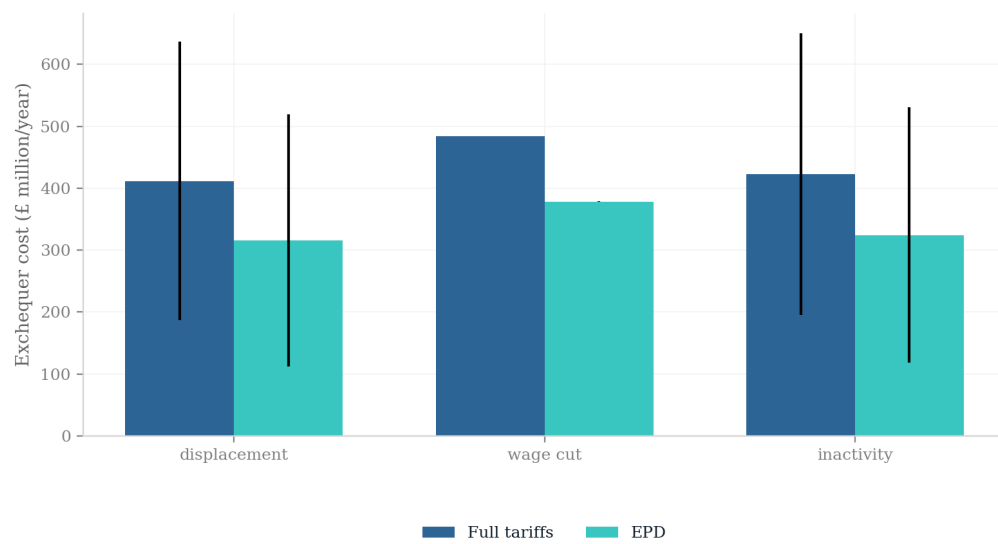


Figure 5: Full-schedule and EPD scenario comparison. Error bars are assignment standard deviations, not causal confidence intervals.

not reported.

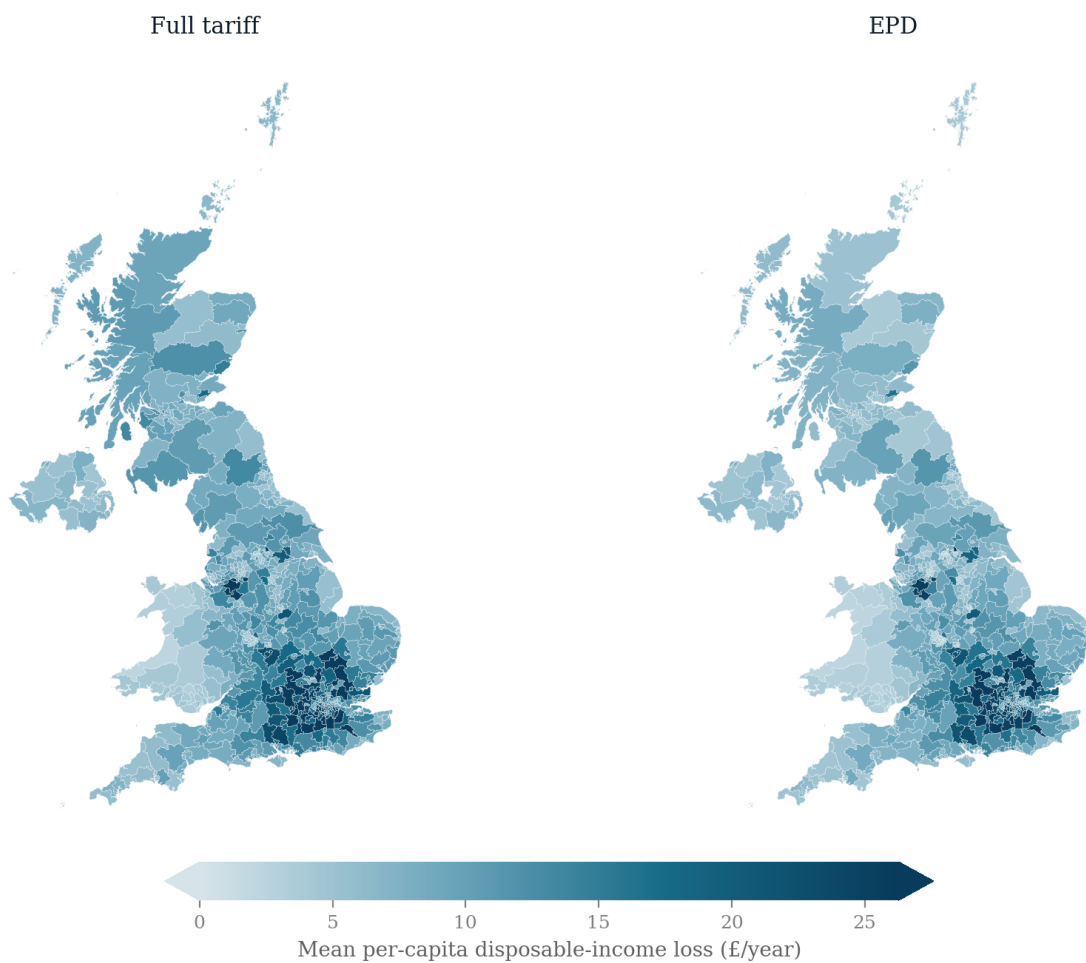


Figure 6: Synthetic constituency illustration of annual per-capita disposable-income loss, averaged over 20 paired worker-assignment draws. A shared sequential scale is clipped at the 95th percentile. The map uses imputed SIC and no observed constituency industry mix; it is not a map of local tariff exposure.

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